Multi-Asset FX Sensitivity & Strategy Dashboard

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Abstract

This report presents a cross-asset macro framework for analyzing major G10 FX pairs using rolling ordinary least squares regressions. The project models currency returns as a function of macro factors spanning interest rates, volatility, commodities, and equity markets. Estimated factor sensitivities are then translated into a macro-aware signal framework and evaluated using a simple strategy backtest. Final outputs include rolling exposure charts, cross-sectional factor comparisons, macro score rankings, and cumulative strategy performance.

1 Introduction

Foreign exchange is inherently a cross-asset market. Currency pairs are often influenced by relative rate expectations, changes in risk sentiment, commodity dynamics, and broader macro regimes. The objective of this project is to construct a quantitative framework that captures these relationships through time-varying factor exposure estimation.

The workflow combines:

- macro factor construction,
- rolling OLS estimation,
- exposure-based signal generation,
- and strategy evaluation.

2 Data Construction

The project uses a macro factor set spanning:

- US 2Y and US 10Y rate changes,
- Germany and Japan 10Y rate changes,
- changes in VIX and MOVE,
- S&P 500, Euro Stoxx, and Nikkei returns,
- WTI, gold, and copper returns.

The FX pairs analyzed are:

- EURUSD
- GBPUSD
- USDJPY
- AUDUSD
- USDCAD

3 Modeling Approach

For each FX pair, a 63-day rolling OLS regression is estimated:

$$r_t^{FX} = \alpha + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \cdots + \beta_k X_{k,t} + \varepsilon_t$$

This rolling framework allows estimated exposures to evolve over time rather than assuming fixed macro relationships.

4 Rolling Factor Sensitivities

Figures below show rolling sensitivities for selected headline factors:

- US 2Y change
- VIX change
- WTI return
- Copper return

5 Cross-Sectional Exposure Analysis

The latest estimated factor exposures are compared across FX pairs.

6 Macro Score and Signal Construction

A simple macro score is constructed by multiplying latest rolling exposures by recent factor direction. A positive score indicates alignment with recent macro moves, while a negative score indicates adverse macro alignment.

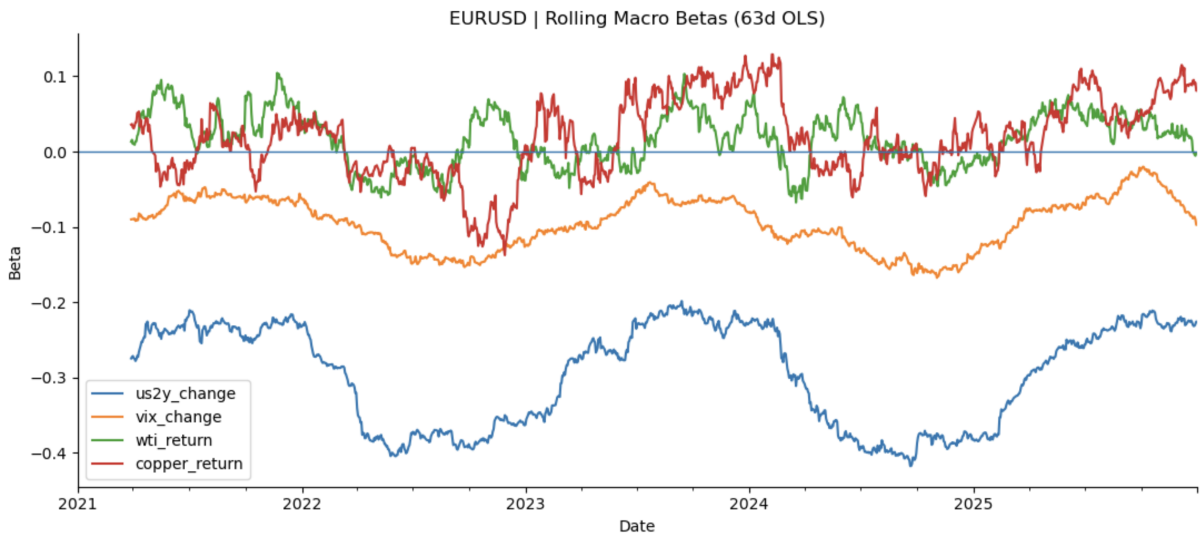

Figure 1: EURUSD rolling macro betas.

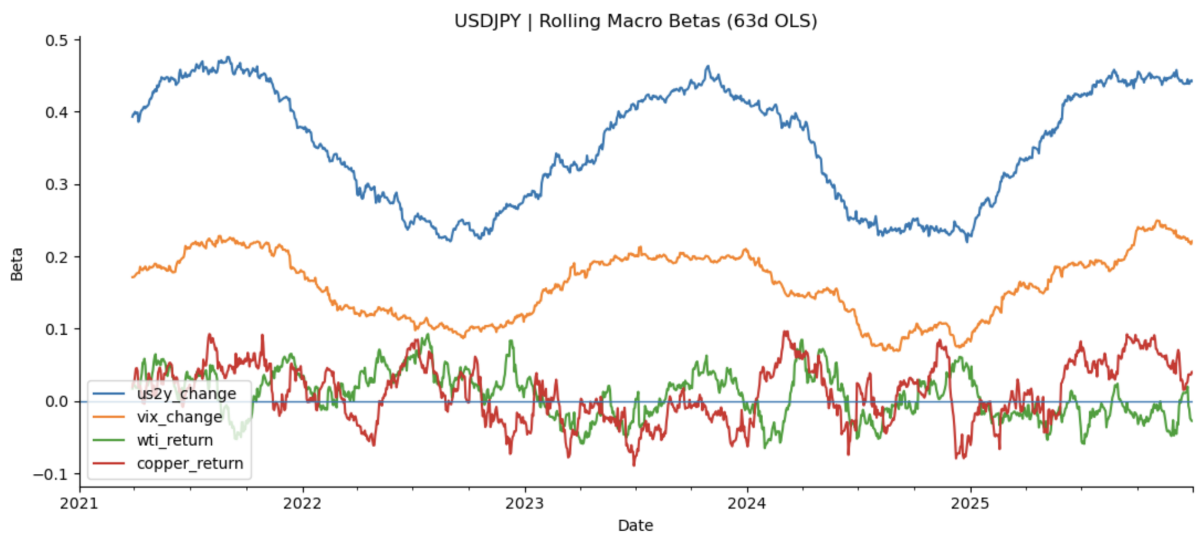

Figure 2: USDJPY rolling macro betas.

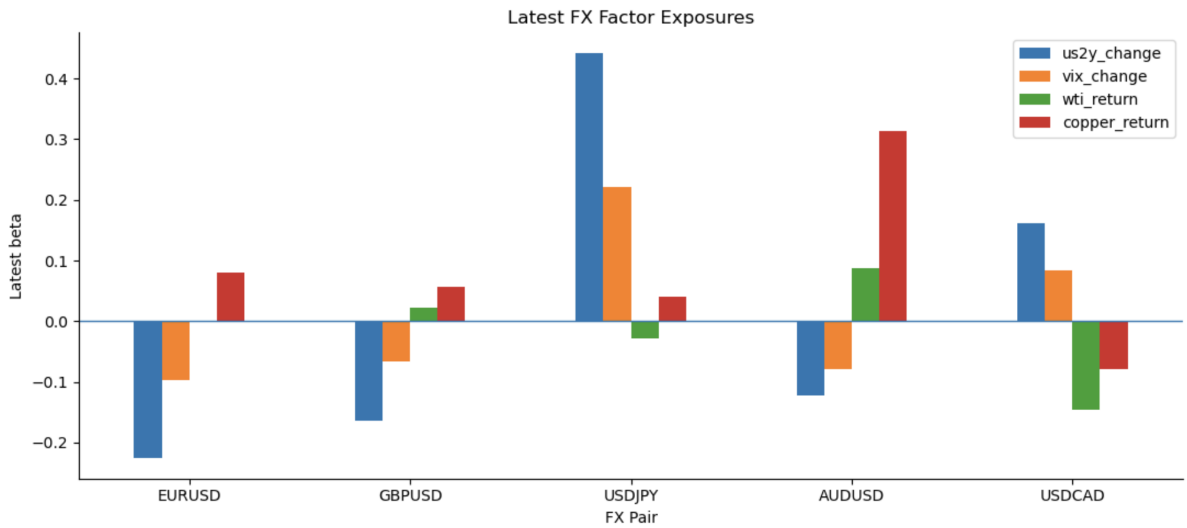

Figure 3: Latest FX factor exposures across major currency pairs.

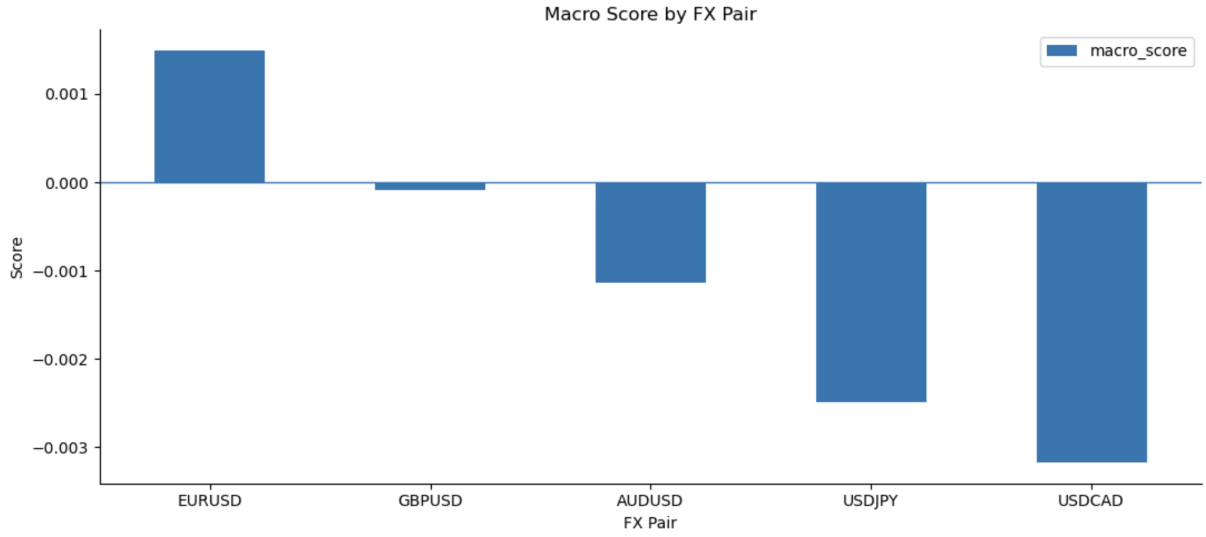

Figure 4: Macro score by FX pair.

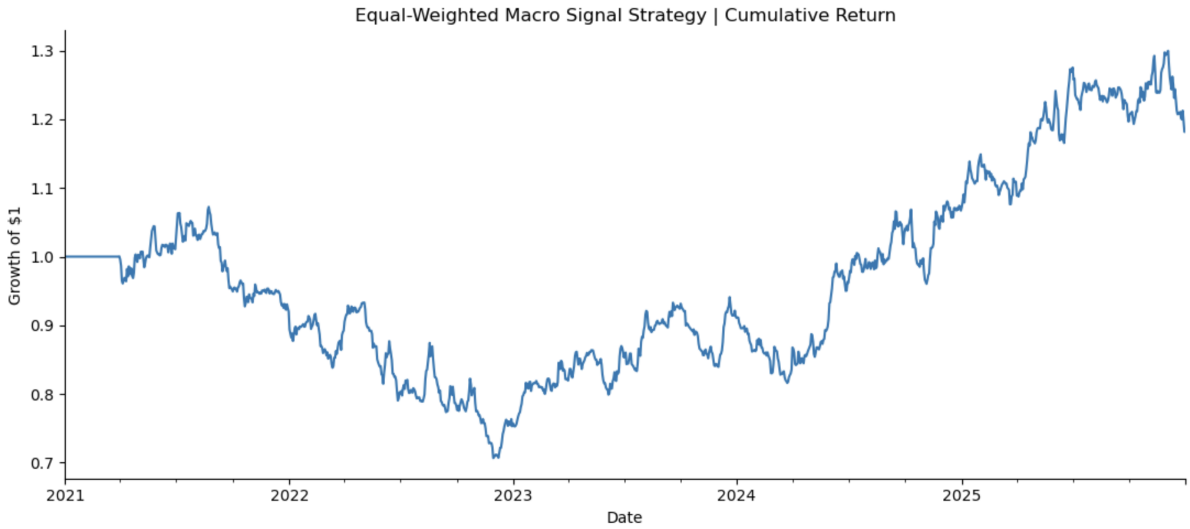

Figure 5: Equal-weighted macro signal strategy cumulative return.

7 Strategy Backtest

Directional trading signals are created using the macro score:

- long if macro score is sufficiently positive,
- short if macro score is sufficiently negative,
- flat otherwise.

The backtest evaluates performance using annualized return, annualized volatility, Sharpe ratio, hit rate, and maximum drawdown.

8 Charts of the Week

To simulate a sell-side or strategy research workflow, the project generates short-form "Charts of the Week" highlighting the strongest and weakest macro setups.

9 Conclusion

This project demonstrates a full quantitative research pipeline for FX macro analysis:

1. building a multi-asset macro factor framework,
2. estimating rolling factor sensitivities,
3. translating exposures into a signal framework,
4. and evaluating results through a simple backtest.

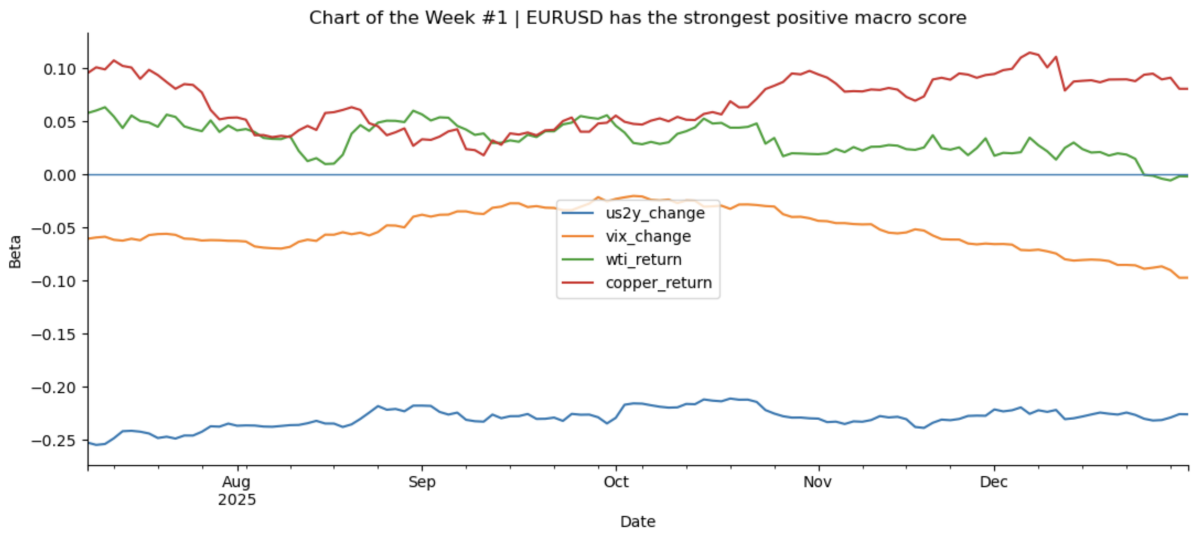

Figure 6: Chart of the Week 1.

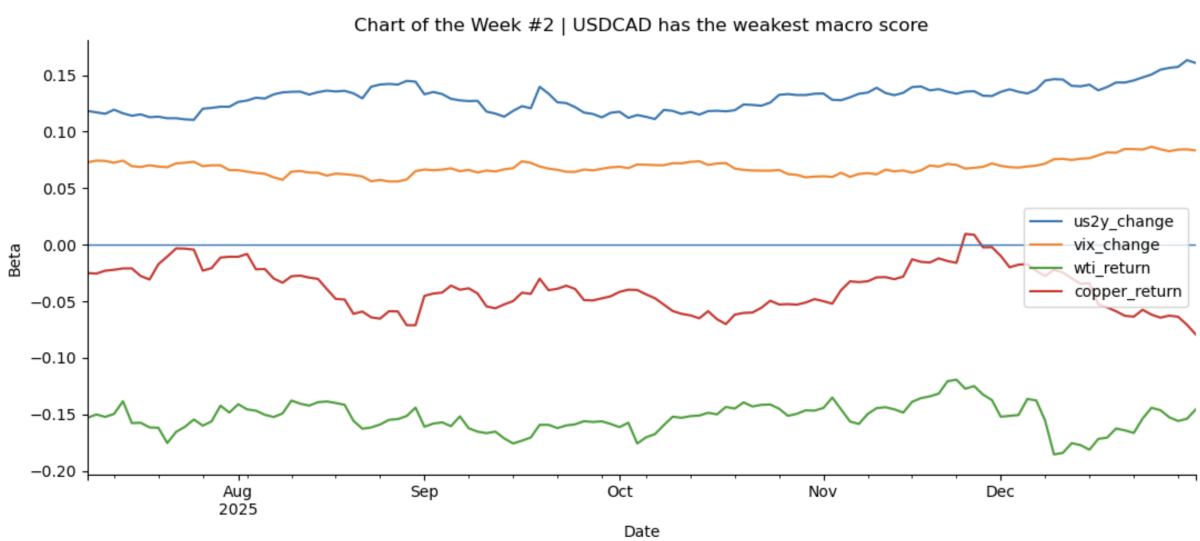

Figure 7: Chart of the Week 2.

The equal-weighted macro signal strategy produced:

- Annualized return: 15.2%
- Annualized volatility: 21.5%
- Sharpe ratio: 0.76
- Maximum drawdown: -12.4%
- Hit rate: 54%

These results suggest that cross-asset macro exposures contain useful directional information for FX markets and can be translated into a systematic signal framework.

The framework is useful as a research prototype and can be extended with real market data, volatility targeting, regime detection, and more advanced factor models.

10 Future Work

Potential extensions include:

- replacing synthetic data with real data feeds,
- adding transaction costs and turnover analysis,
- using PCA or clustering for factor reduction,
- implementing volatility targeting,
- and deploying the framework as an interactive dashboard.